

SLIDE 1

Problem & why it matters

Team name: HawkerBridge Project owner: Shivam Gupta

Problem statement: C3 - KopilamAI: hawker culture and food access

The problem: A closure notice does not tell a community coordinator which neighbourhoods need temporary meal support, or how to use a limited budget.

Who is affected: Residents who rely on nearby hawker food, especially seniors. On 28 September, 17 listed centres with 1,025 food stalls have scheduled closures. These are infrastructure counts, not measured hungry residents.

Why now: Cleaning and renovation closures are known ahead of time. Town councils can compare cleaning dates and prepare support before a neighbourhood loses access.

SLIDE 2

Solution & data

The solution: A closure continuity desk that compares dates and proposes a budgeted, reviewable meal-support plan for neighbourhoods flagged by an access model.

The user: A town council planner or community meal coordinator chooses a date, planning area and budget, checks alternatives, then exports a plan for human review.

Data from data.gov.sg: NEA closure dates, centre locations and stall counts (2026). SingStat Census 2020 population by subzone and age, joined to URA 2019 boundaries. NEA annual food-waste data supplies national context only, never local demand labels.

What differs: Counterfactual cleaning-date comparisons and constrained allocation, with explicit costs, capacity, source provenance and uncertainty. Every allocation has an explicit objective and a comparison baseline.

SLIDE 3

Databricks architecture & impact

Working platform: Public APIs / Lakeflow Job / Bronze raw Delta / Silver validated closures and population / Gold access model and allocation / Databricks Apps. Unity Catalog governs tables. MLflow records the optimiser's baseline comparisons.

Demo: hawkerbridge-sg.web.app offers private plans and PDF export. A live Databricks job passed 12 quality gates and recorded 9 MLflow scenarios, with a SQL dashboard.

Impact to test: Fewer coordinator hours per closure and fewer residents reporting lost meal access. Pilot targets: reduce planning time by 30% and verify every proposed site. A funded pilot and organisation subscription are commercial hypotheses, not sales.

Next sprint: Validate locations with a coordinator, measure planning time and repeat use, then test a paid pilot. Current prices and impact targets are hypotheses.